CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 10: **Light: Mirrors and Lenses** | Quick revision, one page

Laws of reflection

- First law: the incident ray, reflected ray and normal lie in the same plane.
- Second law: $□ngle i = □ngle r$ (both measured from the normal).
- A ray along the normal has $□ngle i = 0^\circ$ and is reflected back along itself.

Plane mirror

- Image is virtual, erect, same size, laterally inverted.
- Image distance = object distance (image is as far behind the mirror as the object is in front).
- Distance between object and image $= 2times$ distance of object from mirror.

Types of reflection

- Regular reflection: smooth surface, parallel rays stay parallel.
- Diffuse reflection: rough surface, reflected rays go in different directions.
- Real image: can be obtained on a screen. Virtual image: cannot.

Spherical mirrors and lenses

- Concave mirror: converging; used as shaving mirror, torch reflector, by dentists.
- Convex mirror: diverging, wide field of view, erect smaller image; used as rear-view mirror.
- Convex lens: thicker at centre, converges rays (magnifying glass, camera). Concave lens: thinner at centre, diverges rays (corrects myopia).

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